

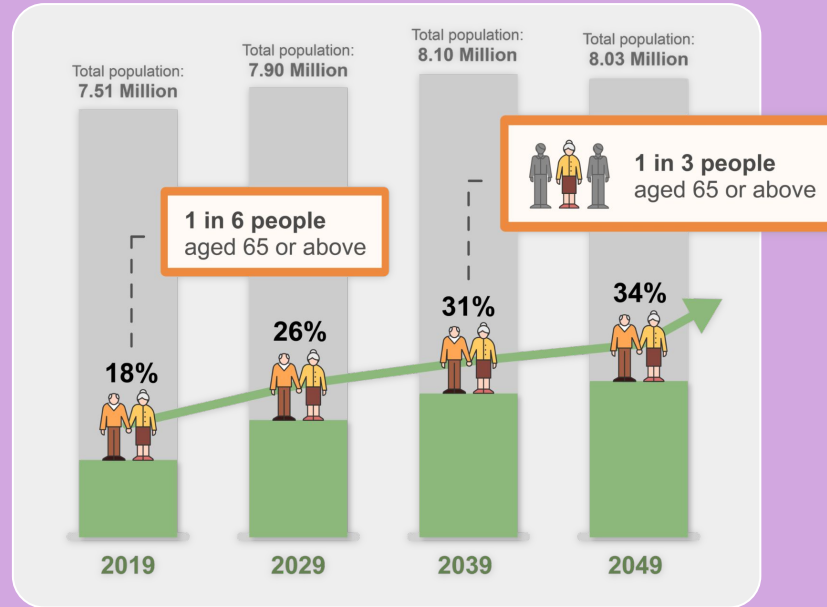
ALICE

Alzheimer Diagnosis and Severity
Indicator based on Eye Behaviour Neural
Network Classifier



Designed & Created by:
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Problem – Aging population in HK with Alzheimer's

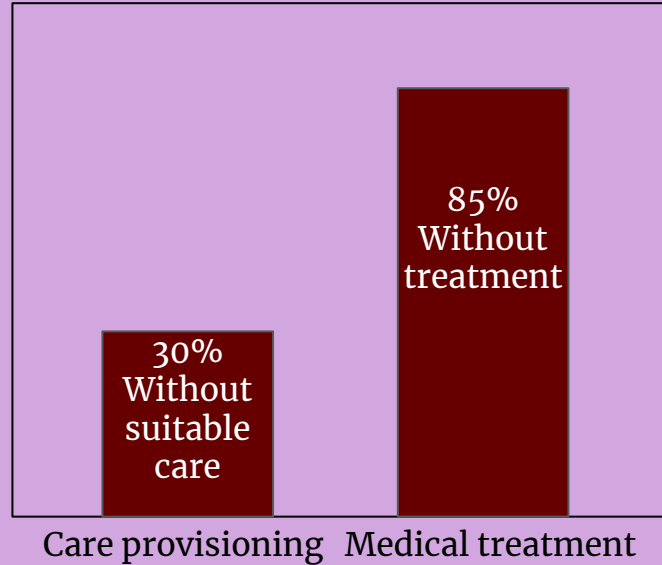


Alzheimer's Disease is a disability that impairs cognitive-communication abilities, posing challenges for care providers in determining appropriate amount of aid for caring for the elderly's wellbeing.

As a rising problem in Hong Kong, by 2049, 1 in 3 citizen* above 65 will have Alzheimer's, adding a heavy cost for society, especially care provider organizations.

Problem – Highly inefficient and wasteful care provision

Elderly
affected with
Alzheimer's



Most elderly with Alzheimer's are not diagnosed nor have medical treatment
They suffer physically and psychologically as they become unable to live independently
Care-providing organizations don't know the level of assistance each elderly needs
Leading to resources and manpower wasted and inefficient allocation.
Results in low efficiency and effectiveness of care provisioning

Our vision – ALICE

Our vision is to enable every elderly with Alzheimer's Disease to receive the perfect degree of care and aid needed to ensure well-being and without interference towards their lifestyles.

Thus, care-provisioning organizations can improve the efficiency and effectiveness by avoiding waste and mis-allocation of resources from neglectance or excessive care.

Goals to achieve to relief the problem:

1. Improve **efficiency** of resource and manpower allocation in care provisioning
2. Improve **effectiveness** of assisting elderly in living independently



Our Solution – ALICE

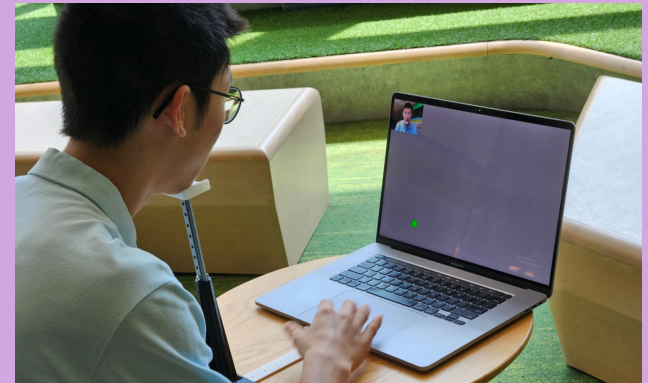
Predict and monitor Alzheimer's severity in elderly up to five years in advance before symptoms hinder wellbeing

Functions - ALICE can predict and indicate the severity of Alzheimer's up to five years in advance, and can track its progression.

Users - The solution is applicable for all ages, however is more advantageous towards elderly with Alzheimer's.

Usage - The user can take the test using any device with an internet connection and a camera, as the test only depends on a mouse and follow instructed eye movements.

Cost - Free for personal use; commercial and organizational usage will require a subscription. Can be very flexible and customizable for each care organization for optimal effect



ALICE – Benefits

AAA

All and Always



Accurate

Organizations can base treatments on the data to give precise aid quickly

1. Doctors see the effectiveness of treatment/prevention
2. Caregivers see the right amount of care needed
3. No over/under allocation of care, reduces inefficiency



Accessible

Most accessible service aimed to help everyone in need

1. Publicly available through internet to reach a wider audience
2. Accessible with minimal equipment, a webcam and internet
3. Remote access to conduct diagnosis anywhere to provide maximum coverage



Accelerated

Elderly can frequently take the test for most timely data

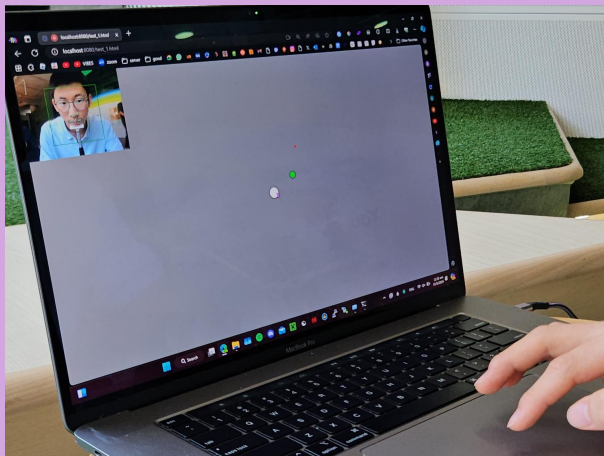
1. Tracks the progression of their conditions
2. Provide doctors and caregivers most timely data
3. Understands their current situation to give effective care

ALICE - Demo



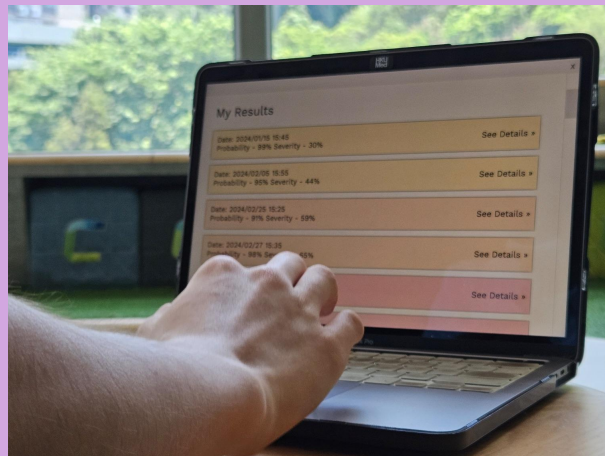
Accessible

ALICE is accessible on any device with internet access and a camera, e.g. smart phone, tablet, laptop



Accurate

ALICE can produce results instantly after testing, and each test will require less than 5 minutes



Accelerated

ALICE can track disease progression over time to help caregivers and doctors assess elderly long-term

ALICE – Scientific principle

Prior research proved Alzheimer elderly respond differently to unaffected people when exposed to visual stimulation such as color, movement, and brightness.

We combine five tests to record and evaluate the behaviour of user's saccadic functions, eye motion, blinking count, and more. The user's behaviour is compared to that of a normal person or an Alzheimer's elderly to determine the user's conditions & severity.

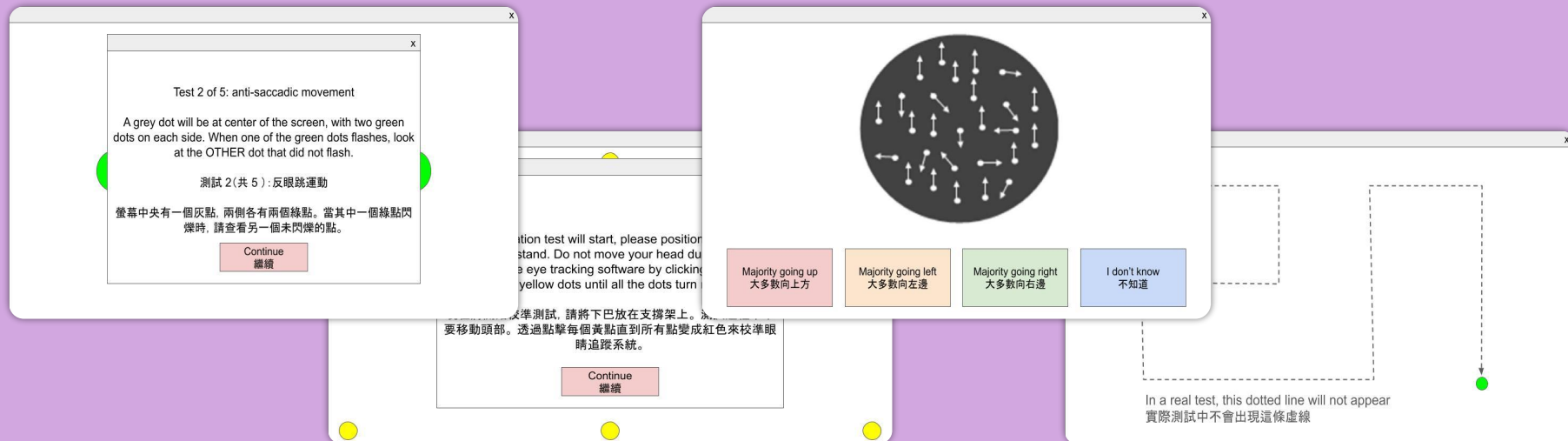


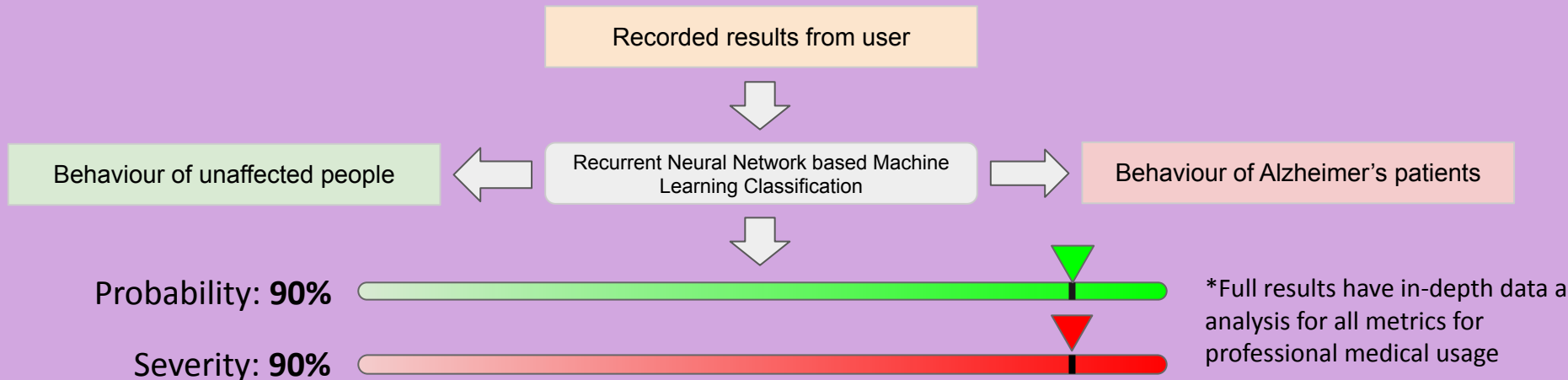
Fig 1-4 . ALICE Diagnostic website interface

ALICE – Scientific Innovation

First ever to apply research through fully automated software (Behaviour observation)

First ever software to diagnose Alzheimer's only based on VISUAL Functions & Features

Uses a Recurrent Neural Network Machine Learning classification based on 400 AD patients' data



Based on the results similarity to the patient, we can deduce the probability of an user having Alzheimer's as well as its severity.

ALICE – Future Development

1. Expand Users base (non-commercial) FREE USAGE - Promotions, Incentives,

- Create incentives by offering **FREE** plan for TRIAL usage period, as well as reaching out to elderly homes and other care-provisioning organizations to acquire as much elderly users as possible.
- Non-commercial **ACADEMIC EXCHANGE** with Hong Kong universities and research institutions around the world to contribute to scientific research and benefit more people in the future.

2. Business Collaboration with service provisioning organizations - Commercialization steps

- Negotiate subscription based plans for service provision organizations based in HK, such as elderly homes and hospitals in Hong Kong SAR.
- Extend services to mainland CHINA and other regions by adapting ALICE to fit foreign medical and ethical protocols as well as obtain connection with foreign elderly homes and hospitals.

3. Long-term improvements - Scientific research & development

- Expand dataset of different Alzheimer patients behaviours to **CONTINUOUSLY** improve on the accuracy of the model. User input and self-reports of diagnostic tests will also help improve the accuracy of the model.
- Active development of integrating more sub-tests and test metrics and the most up-to-date scientific support. Offer a longer test that gives more holistic and in-depth analysis of conditions.